BABU BANARASI DAS UNIVERSITY

SESSION 2026–27

TELECOMMUNICATIONS CUSTOMER CHURN PREDICTION

PROJECT – 1 | CHAID-Based Customer Churn Prediction

A Data Mining Project Using IBM SPSS Modeler 18.0

SUBMITTED BY	SUBMITTED TO
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Predictive Analytics • Customer Churn • IBM SPSS Modeler

PROJECT OVERVIEW

Executive Summary

Customer churn occurs when a customer stops using a company's service. This project documents a predictive-analytics workflow for identifying telecommunications customers who may be likely to leave. Historical customer records are inspected and filtered, a CHAID decision-tree model is developed in IBM SPSS Modeler, and the trained model is applied to current-customer records. Customers predicted as churners with confidence greater than 0.90 are selected for follow-up by the retention team.

Business Problem

Telecommunications providers need to identify customers at risk of discontinuing their service before the customer leaves. Reviewing every customer manually is inefficient. A predictive model can help identify patterns in customer and service attributes and prioritize retention efforts for the customers who may need attention first.

Project Aim

To demonstrate an end-to-end CHAID-based customer churn prediction workflow, from data inspection and preparation through model training, scoring, risk filtering, and export of a customer-level output file.

Objectives

- Inspect historical telecom customer data and understand its fields.
- Remove records marked as unknown using the *data_known* field.
- Assign *churn* as the prediction target and suitable customer/service fields as inputs.
- Train and interpret a CHAID decision-tree model.
- Score current-customer deployment records using the trained model.
- Filter customers predicted as Churned with confidence above 90%.
- Export a clean list containing customer ID, predicted outcome, and churn probability.

TOOLS, DATASET AND PREPARATION

Software

IBM SPSS Modeler 18.0 is the visual data-mining environment referenced in the supplied project material. Its source, filtering, field-role, modeling, and output nodes are connected as a stream so the workflow can be inspected and rerun.

Data files referenced by the demonstration

File / field	Purpose
telco_x_modeling_data.xlsx	Historical customer records used to build the model.
telco_x_deployment_data.xlsx	Current-customer records to which the trained model is applied.
data_known	Used to retain records whose data is known (value: yes).
churn	Target field the model predicts.
customer_id	Identifier retained in the final scored output.

Record counts reported in the supplied material

The demonstration refers to 31,789 historical records before filtering and 31,769 records after retaining records where *data_known* = *yes*. At the root node, 14,713 records are reported as churned, which is approximately 46.3% of the 31,769 records. These are values reported by the source demonstration and have not been independently recalculated from the original Excel files.

Data preparation considerations

- Preview the imported fields and sample records before modeling.
- Exclude records with unknown/null information according to the demonstrated filter.
- Set the churn field as the target and appropriate customer/service attributes as inputs.
- Do not use *retention* as an input if it directly duplicates the churn outcome; this would introduce target leakage.

METHODOLOGY / WORKFLOW

The workflow is organized into the following steps. The accompanying original project pages provide visual evidence and interface-level instructions for the IBM SPSS Modeler process.

Step 1 — Import historical data

Add an Excel source node and select the historical modelling workbook. Preview the fields and sample records to confirm that the data has been loaded correctly.

Step 2 — Inspect data quality

Review the data_known field and identify records marked no, which the demonstration describes as containing undefined or null values.

Step 3 — Filter usable records

Add a Select node and retain records where data_known = yes. The supplied material reports 31,769 records after filtering.

Step 4 — Configure field roles

Use a Type node to inspect field values. Set churn as the target and relevant customer, handset, tariff, usage, and revenue fields as inputs. Exclude leakage fields such as retention.

Step 5 — Train the CHAID model

Build a CHAID decision tree. CHAID selects branches by testing relationships between input fields and the target, creating customer segments with different observed churn patterns.

Step 6 — Interpret the tree

Review the root node and branches. In the supplied example, handset is the first split variable, with tariff and gender appearing in subsequent branches for some segments.

Step 7 — Score deployment records

Import the current-customer workbook, apply the same known-data filter, and connect the trained model to generate predicted churn and confidence fields.

Step 8 — Select high-risk customers

Filter records predicted as Churned with confidence greater than 0.90. Rename the relevant output fields to predicted_value and probability_to_churn.

Step 9 — Export the output

Retain customer_id, predicted_value, and probability_to_churn and export the selected records to a delimited text file such as customers_at_risk.csv.

FINDINGS, OUTPUT AND CONCLUSION

Key findings reported by the demonstration

- 31,769 historical records remain after filtering unknown records.
- 14,713 records are reported as churned at the root node, approximately 46.3% of the modeling records.
- The illustrated decision tree first splits on handset; tariff and gender appear in later branches for particular customer segments.
- The demonstration reports a 94.827% churn percentage for a handset group containing ASAD90, CAS30, SOP10, and SOP20. This source-reported result should be checked against the original data before being used for business decisions.
- The final workflow selects customers predicted to churn with confidence above 90%.

Expected deliverable

A scored customer list containing customer_id, predicted_value, and probability_to_churn. The list is intended to help a retention team prioritize customer outreach. The supplied material describes the workflow but does not include a final exported customer list.

Limitations and recommendations

- The original Excel datasets and runnable SPSS stream were not included with the supplied demonstration PDF, so the reported tree cannot be independently reproduced from the available material.
- No independent model-performance measures—such as accuracy, precision, recall, F1-score, or ROC-AUC—are provided. Evaluate the model on held-out data before operational use.
- Use a confusion matrix to examine false positives and false negatives, and check for class imbalance.
- Confirm that each predictor is available before churn occurs. Treat predicted probabilities as estimates, not guarantees, and monitor the outcomes of retention campaigns.

Conclusion

This project demonstrates an end-to-end churn prediction process: data inspection, filtering, field-role assignment, CHAID model training, tree interpretation, scoring, high-risk selection, and export. It illustrates how predictive analytics can support proactive customer retention. The next step is to run the stream with the original data, validate it on unseen records, and measure whether targeted retention actions reduce actual churn.

Reference

Project workflow and figures adapted from the supplied CHAID-Based Customer Churn Prediction demonstration PDF. Dataset counts and tree findings are reported as shown in that source material.

PREDICTIVE ANALYTICS PROJECT – 1

Project Objective

The objective of this project is to use historical customer information to develop a predictive model that identifies customers who are more likely to discontinue a service. The trained model is then applied to current customer records to identify potential churn cases for further action.

Software Used: IBM SPSS Modeler

Input data:

- demo_data_working_with_modeler.xlsx
- telco x deployment data.xlsx

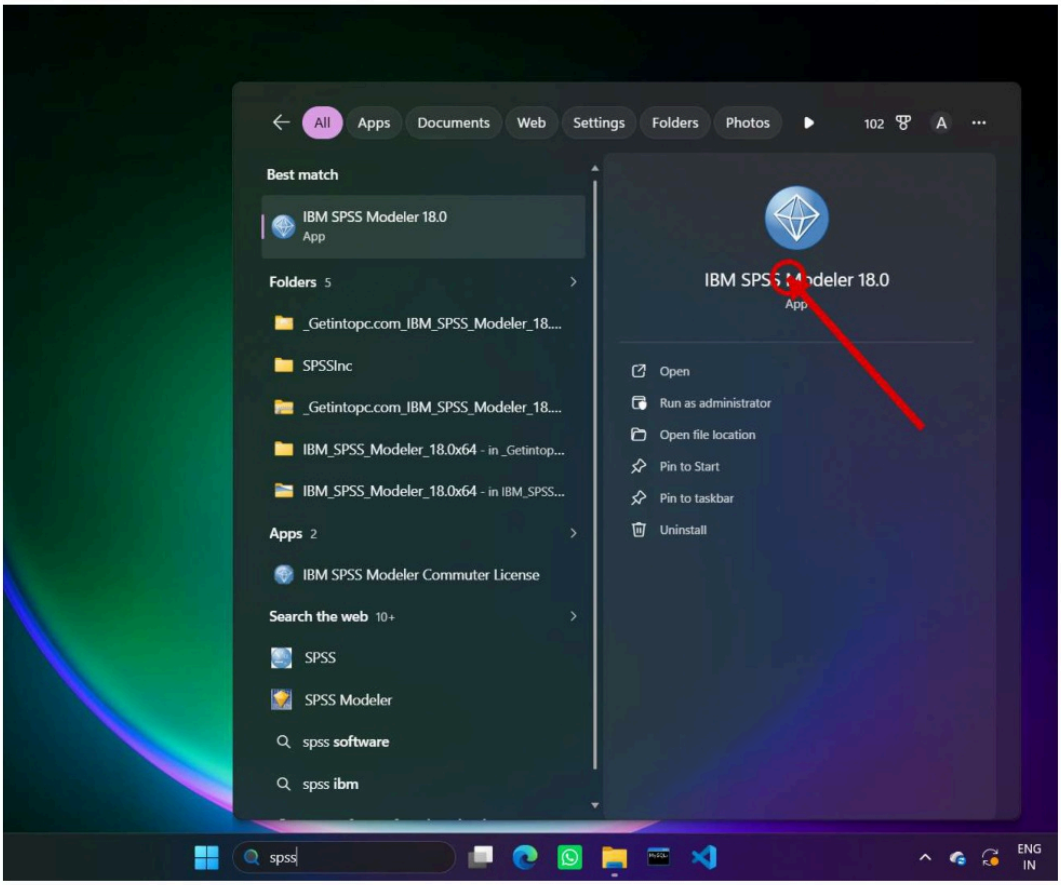
Working folder: C:\Users\vishal\OneDrive\Desktop\Vishal_IBM

Task 1 – Launch IBM SPSS Modeler and set the working directory

What to do: Start IBM SPSS Modeler, create the working stream, and set the project folder so that the input files are available from the correct location.

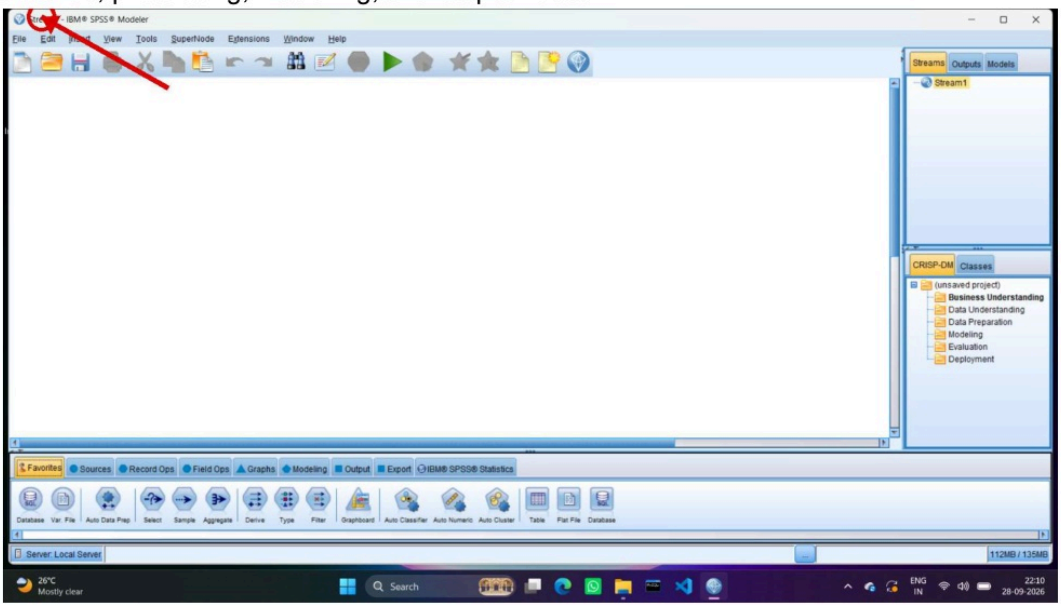
Step 1 – Launch IBM SPSS Modeler

Open the Windows search panel and select IBM SPSS Modeler 18.0. This starts the application and opens the Modeler workspace for creating the predictive analytics stream.


All
Apps
Documents
Web
Settings
Folders
Photos
Best match
IBM SPSS Modeler 18.0
App
Folders 5
_Getintopc.com_IBM_SPSS_Modeler_18....
SPSSInc
_Getintopc.com_IBM_SPSS_Modeler_18....
IBM_SPSS_Modeler_18.0x64 - in _Getintop...
IBM_SPSS_Modeler_18.0x64 - in IBM_SPSS...
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spss software
spss ibm
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Run as administrator
Open file location
Pin to Start
Pin to taskbar
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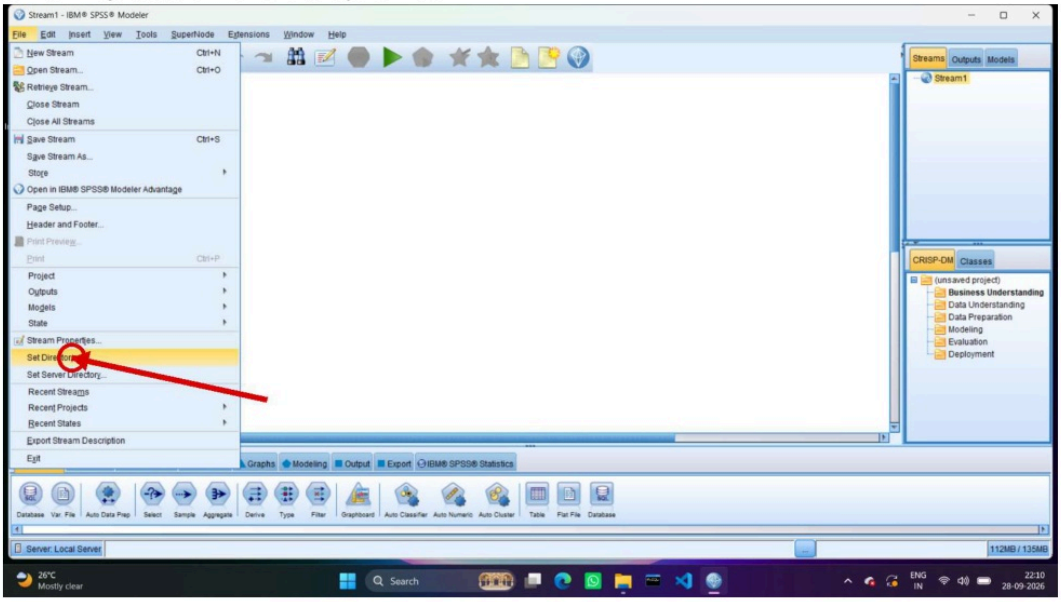
Step 2 – Open the Modeler stream workspace

After the application starts, the main Stream1 workspace is displayed. This blank canvas is used to place source, processing, modelling, and output nodes.



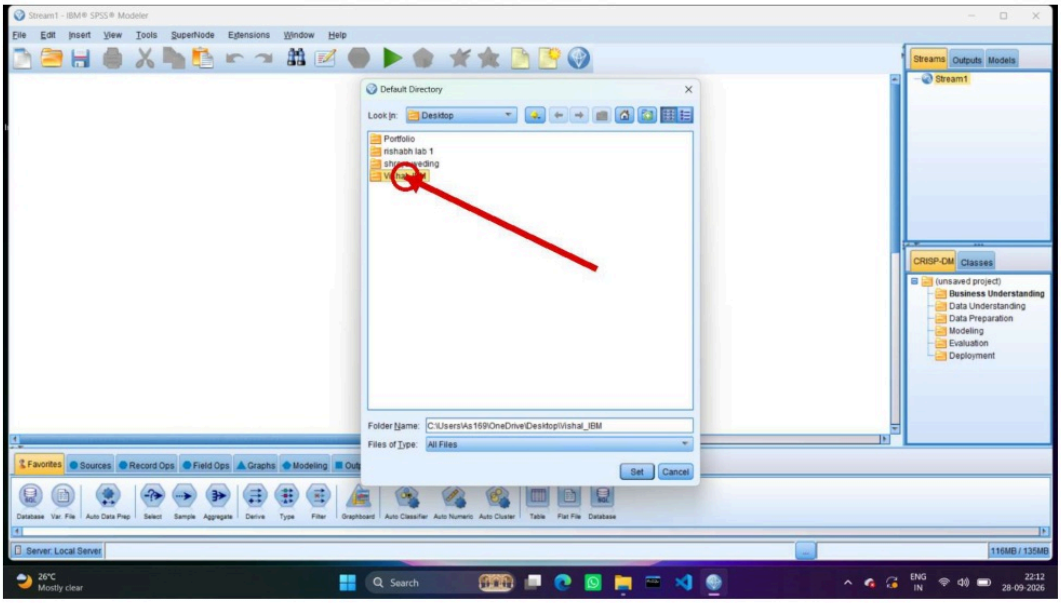
Step 3 – Open Set Directory

Open the File menu and choose Set Directory. This option is used to define the folder that will act as the working location for the project files.



Step 4 – Select the project folder

In the Default Directory window, select the Vishal_IBM folder from the Desktop location. Confirm the folder so Modeler uses it as the working directory.

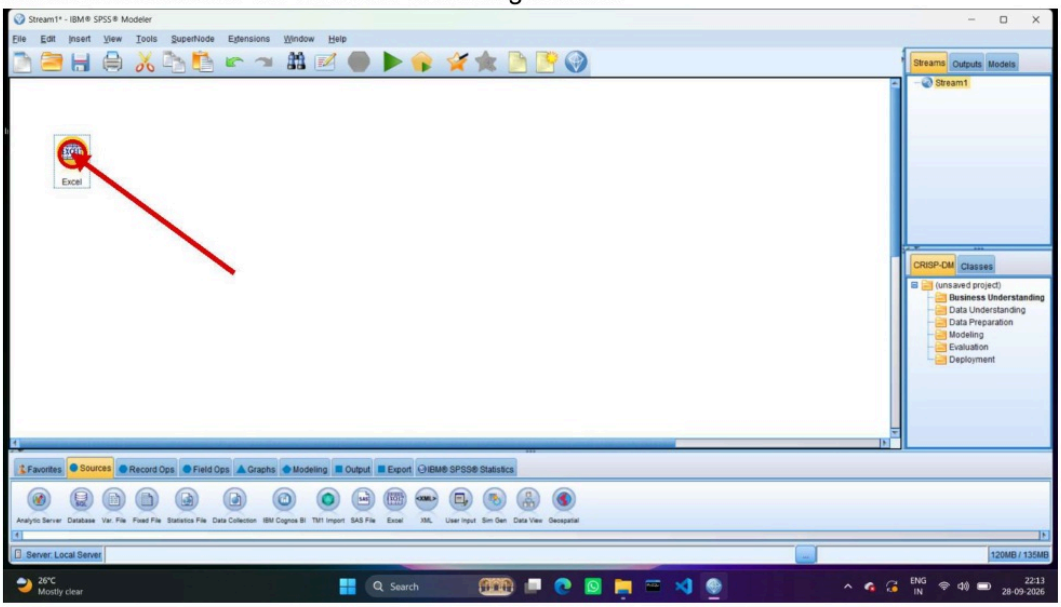


Task 2 – Import and inspect the historical modelling data

What to do: Bring the historical Excel dataset into SPSS Modeler and verify that the customer records have been loaded correctly before further processing.

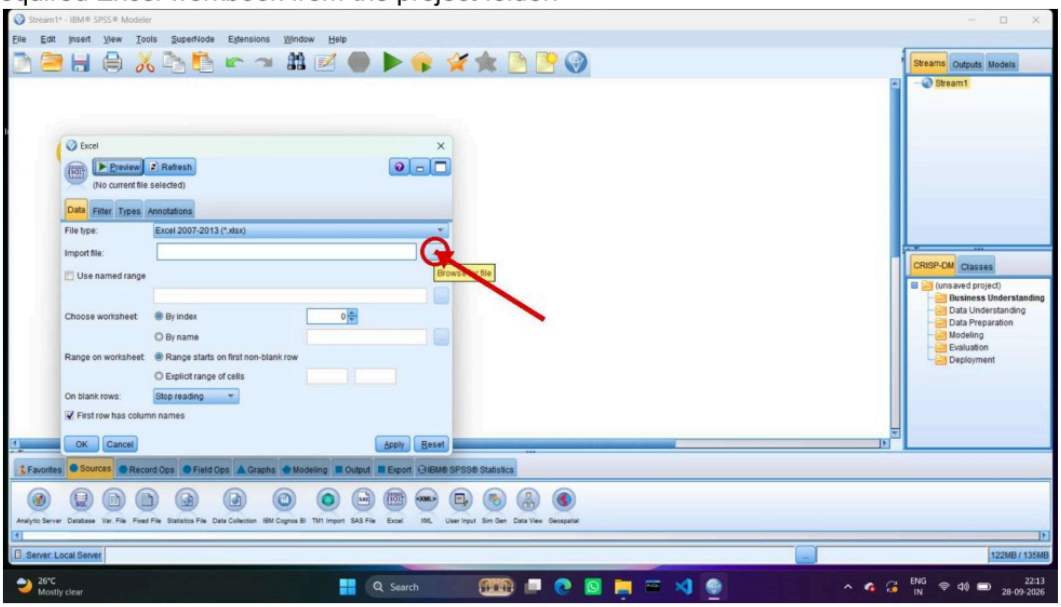
Step 5 – Add an Excel source node

From the Sources palette, select the Excel source node and place it on the stream canvas. This node will be used to read the historical modelling dataset.



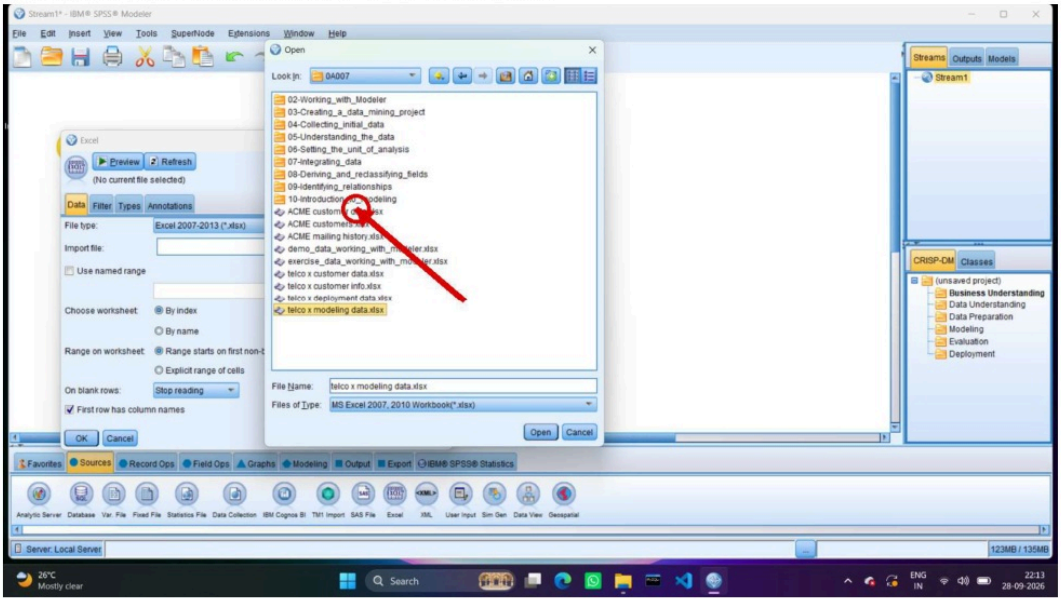
Step 6 – Open the Excel source settings

Open the Excel node and locate the Import file section. The Browse for file control is used to select the required Excel workbook from the project folder.



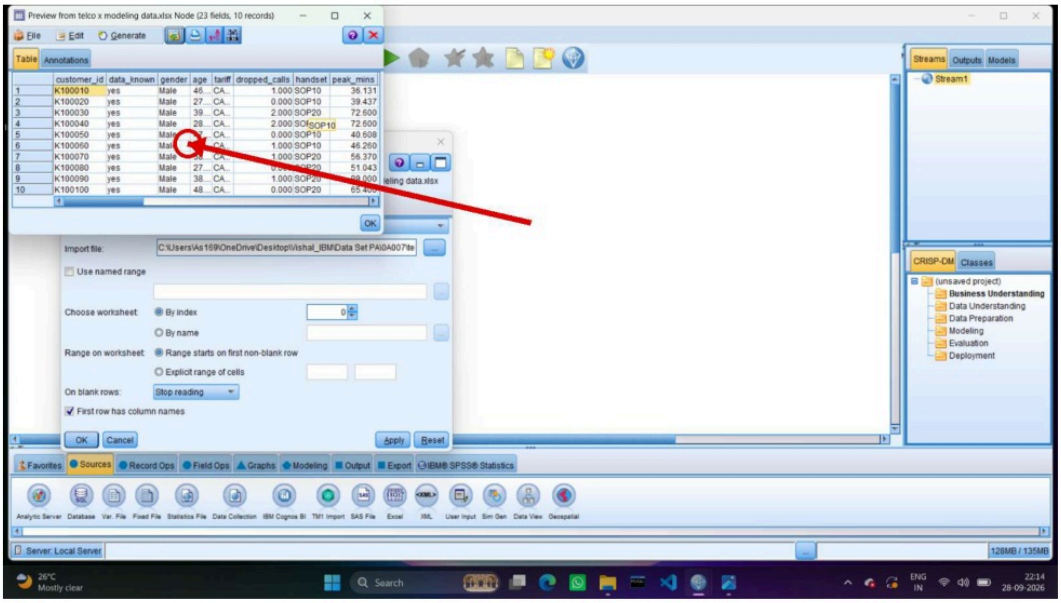
Step 7 – Select the modelling workbook

Browse through the available files and select the telco x modeling data.xlsx workbook. The selected file is then shown as the source for the Excel node.



Step 8 – Preview the imported records

Use the preview to verify that the workbook has been read successfully. The displayed rows and columns confirm that customer attributes such as gender, age, tariff, dropped calls, handset, and peak minutes are available.

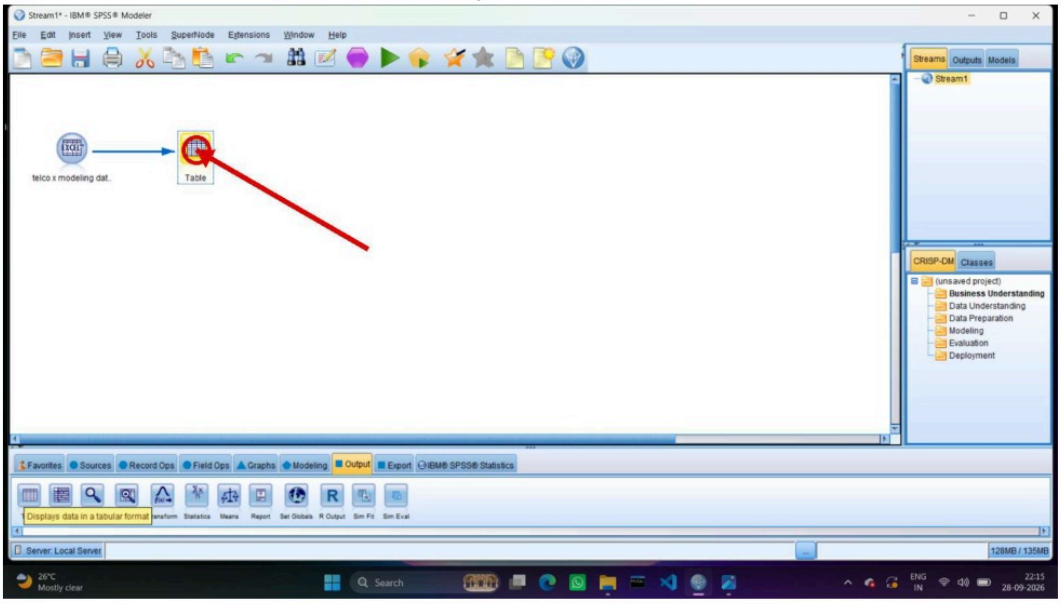


Task 3 – Create a table and generate a field-processing node

What to do: Create a tabular view of the historical data, generate a selection/processing node from the table, and configure its name for later use in the stream.

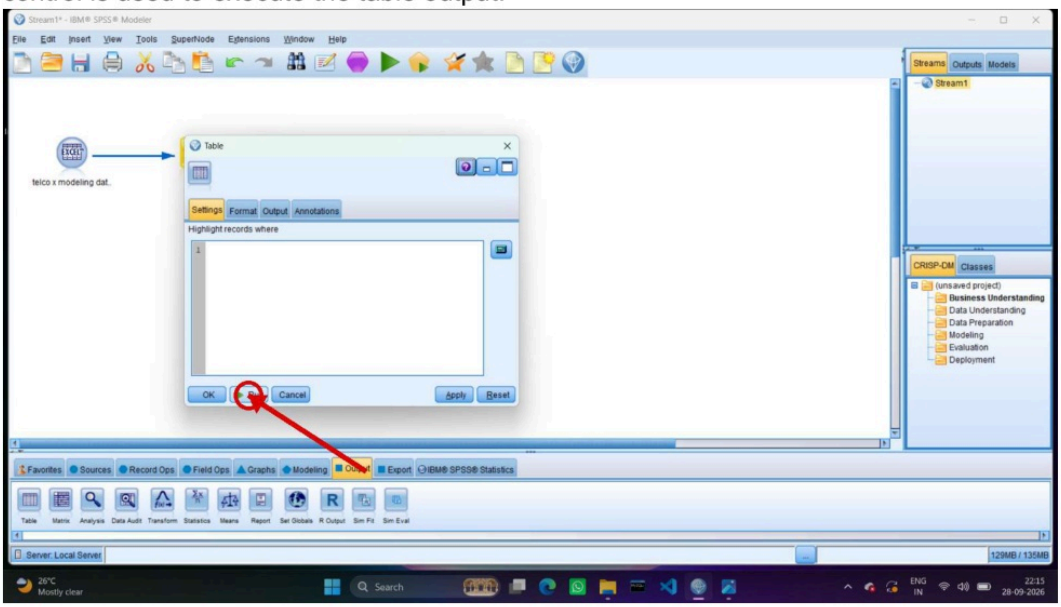
Step 9 – Add and connect a Table node

Place a Table node after the Excel source and connect the two nodes. The table output provides a direct view of the records contained in the imported dataset.



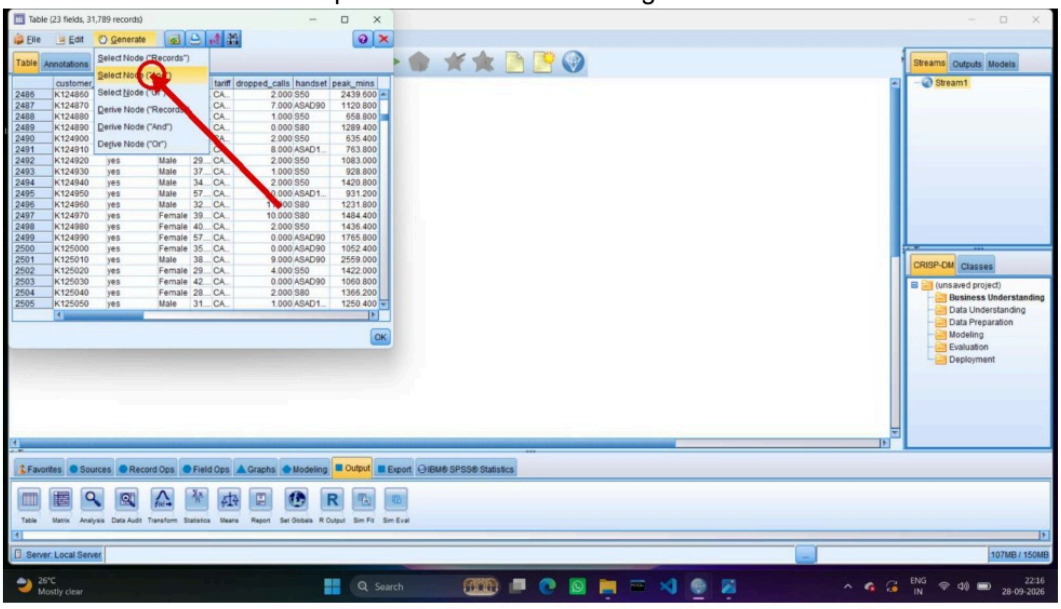
Step 10 – Configure the Table output

Open the Table node and review its settings. The table is configured to display the records, and the Run control is used to execute the table output.



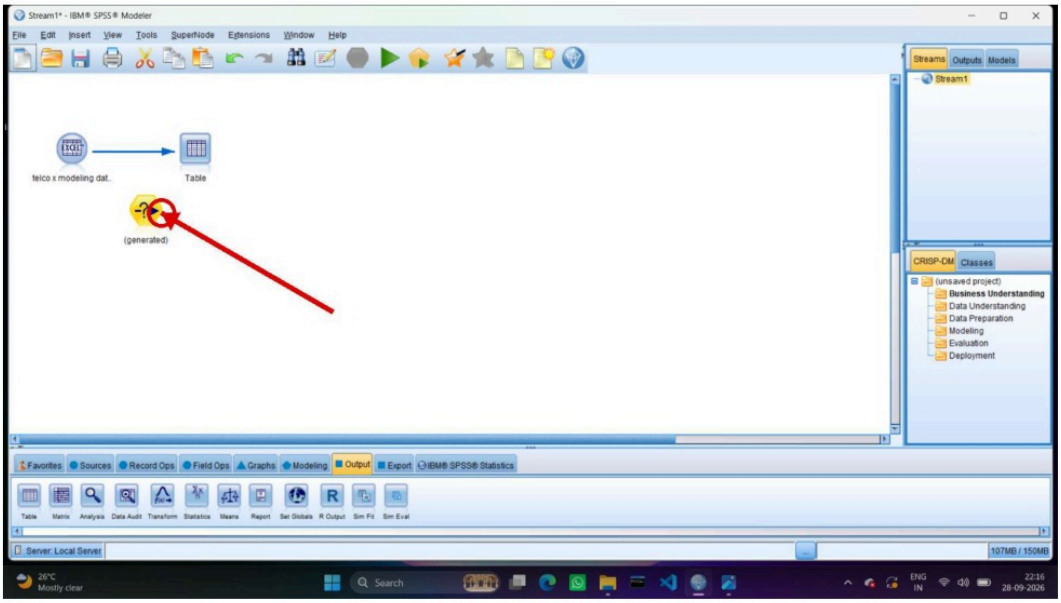
Step 11 – Generate a Select Node

In the generated table, open the Generate menu and choose Select Node ('And'). This creates a selection node from the table output for use in the modelling stream.



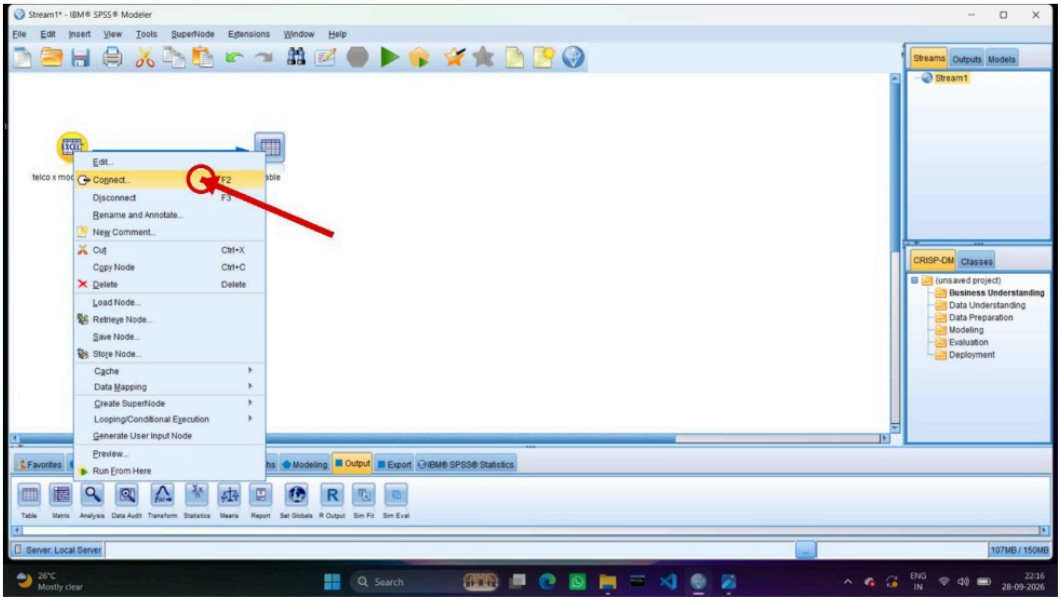
Step 12 – Confirm the generated node

The generated node now appears on the stream and is connected with the table branch. It provides the generated record-selection logic that can be reused in subsequent processing.



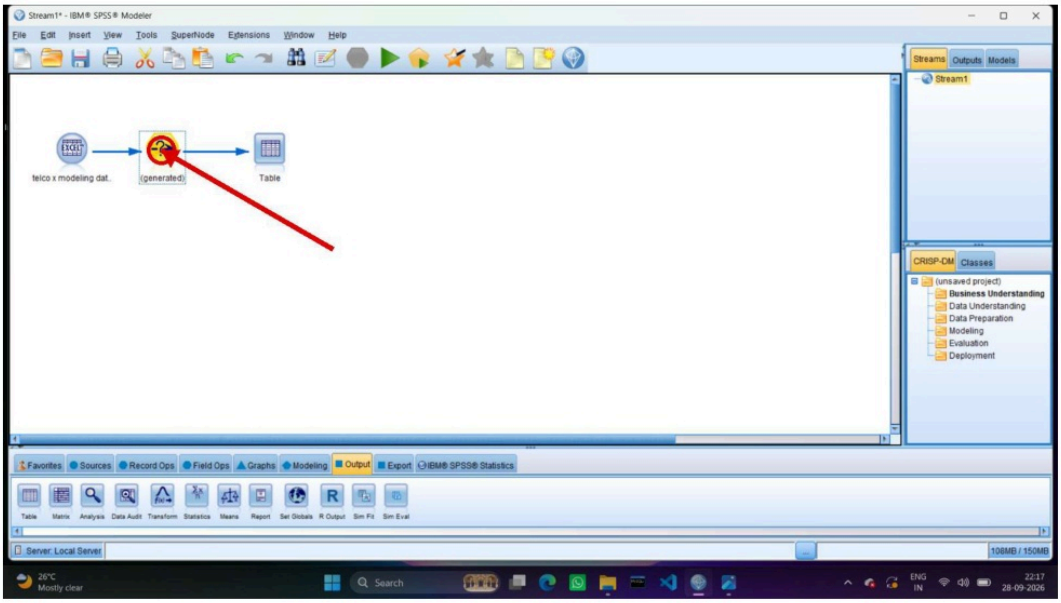
Step 13 – Connect the generated node

Right-click the generated node and choose Connect. This allows the generated node to be linked into the required stream path.



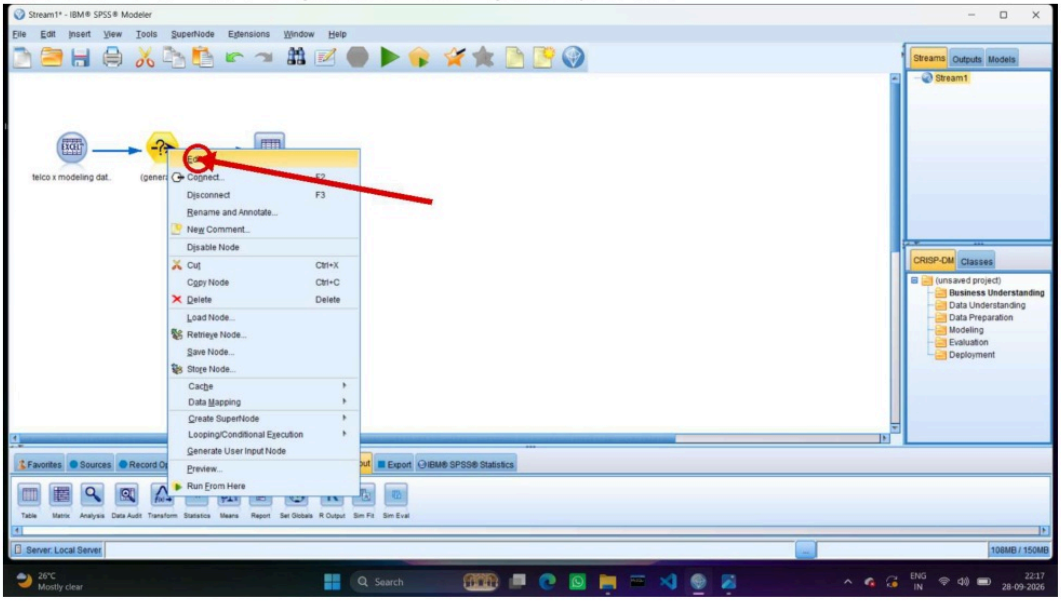
Step 14 – Verify the generated-node connection

The stream now shows the generated node positioned between the source and the Table node. This confirms that the generated operation has been inserted into the processing flow.



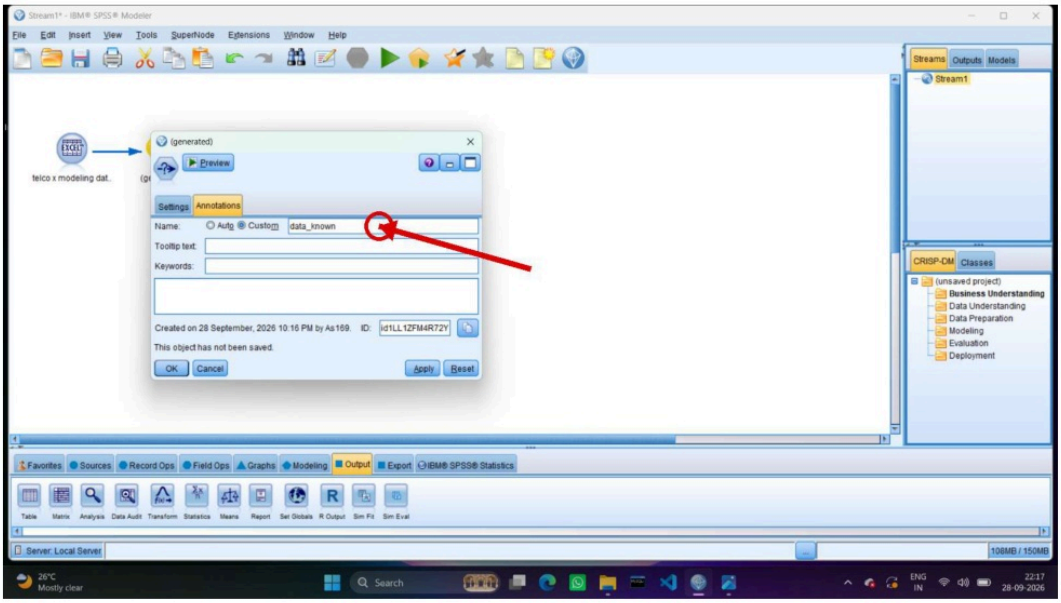
Step 15 – Open the generated node for editing

Right-click the generated node and choose Edit. The node properties are opened so its annotation/name can be changed to a meaningful project label.



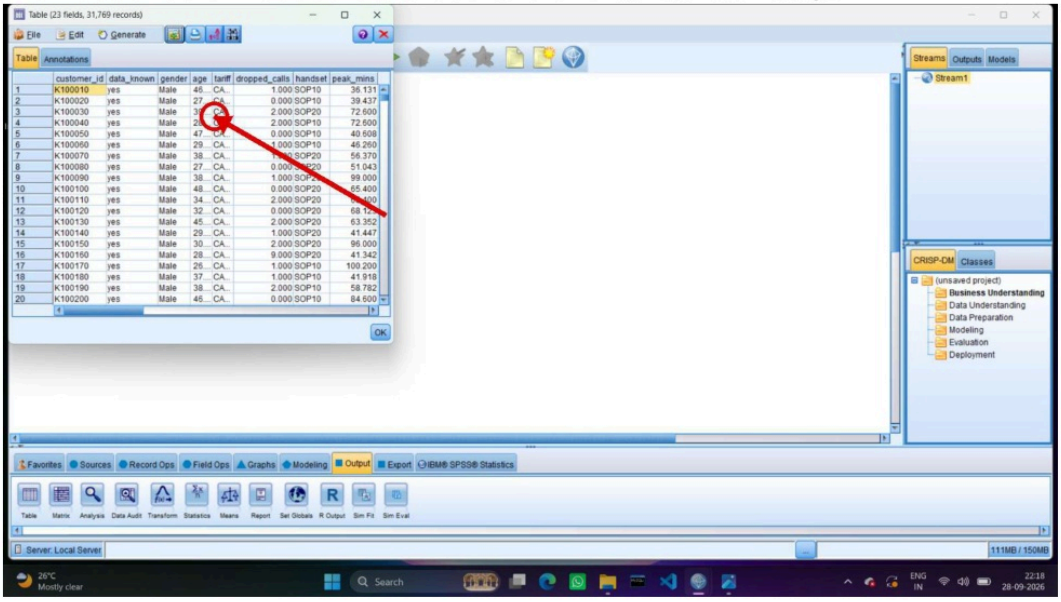
Step 16 – Rename the generated node

In the node settings, select the Custom name option and enter data_known. This name makes the purpose of the generated node easier to identify in the final stream.



Step 17 – Preview the processed records

Use Preview to inspect the records associated with the generated node. The table confirms that the customer fields and their values are available for the next preparation stage.

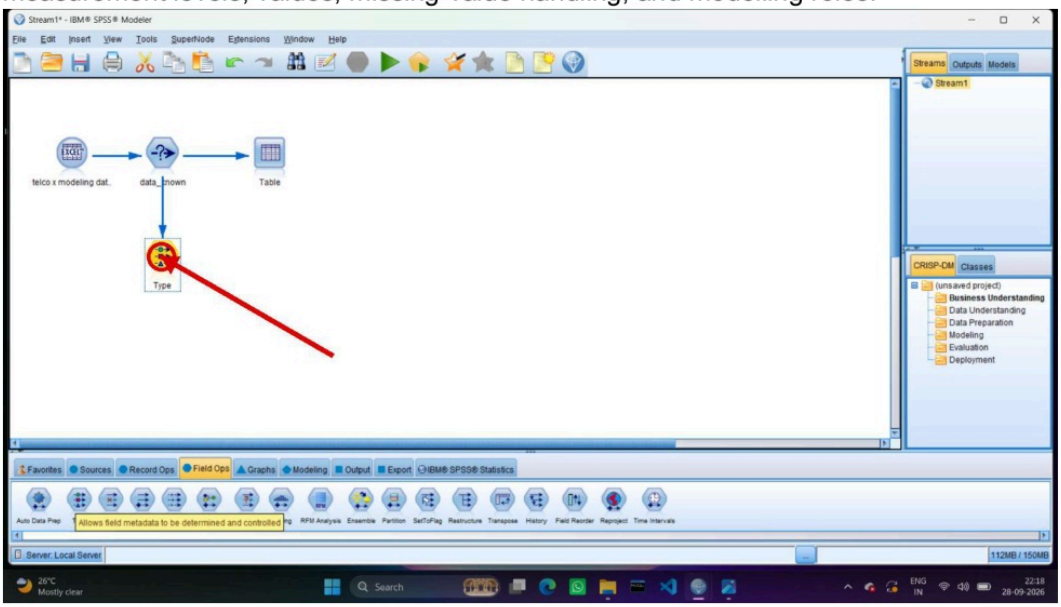


Task 4 – Prepare the data and define field metadata

What to do: Add a Type node and review the measurement levels and field roles so the modelling algorithm receives correctly defined inputs.

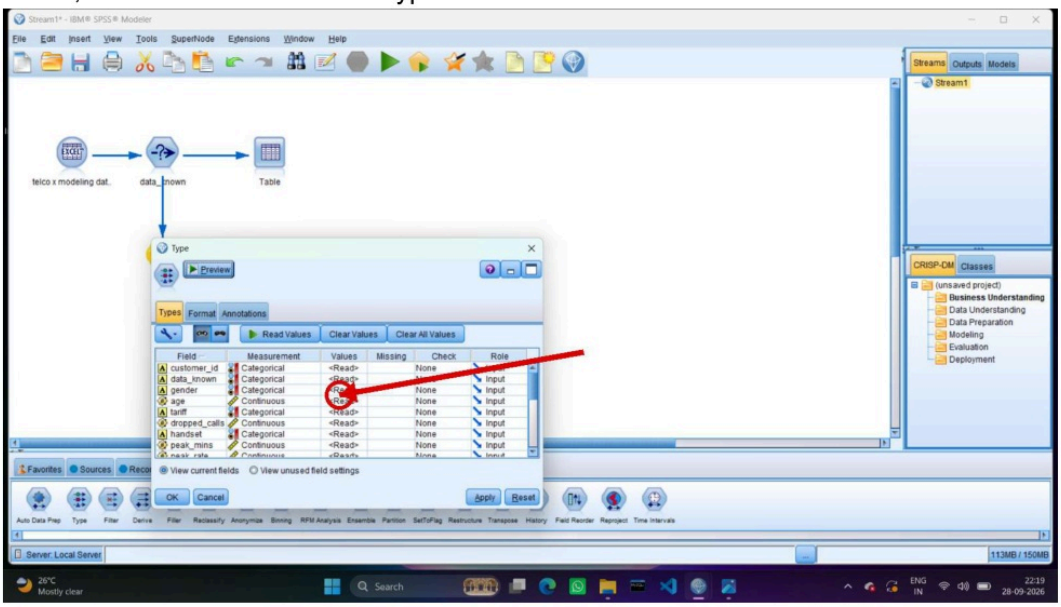
Step 18 – Add the Type node

Place a Type node below the data_known node and connect it to the stream. The Type node controls field measurement levels, values, missing-value handling, and modelling roles.



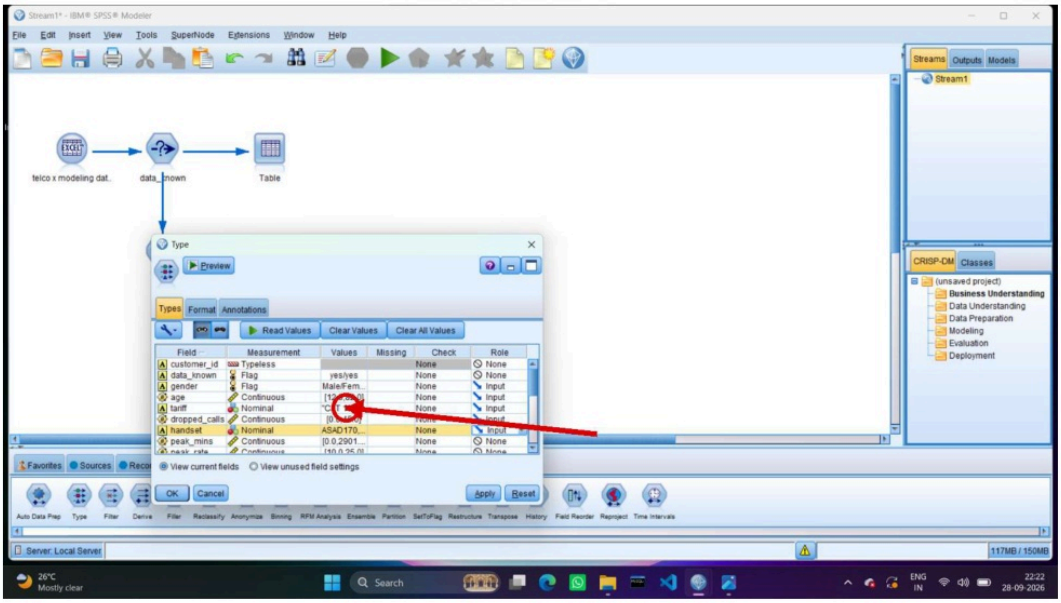
Step 19 – Review field roles in the Type node

Open the Type node and review the listed fields. The displayed customer variables are treated as input fields, while their measurement types are shown beside each field.



Step 20 – Check measurement levels and values

Review the measurement and value information for the imported fields. Confirm that categorical fields such as gender and tariff and continuous fields such as age and dropped_calls are represented appropriately.

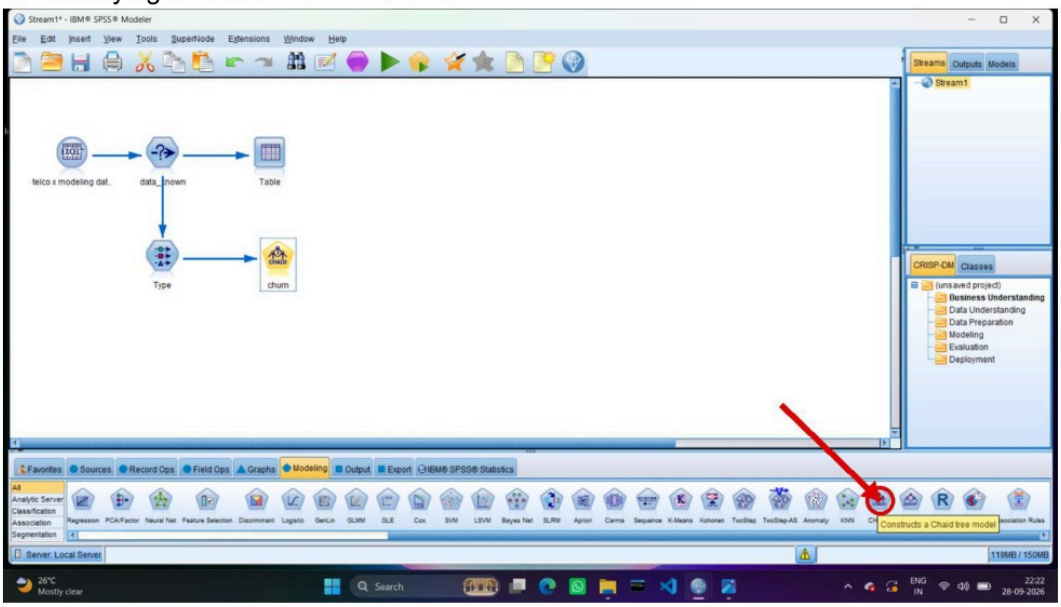


Task 5 – Build and run the CHAID churn model

What to do: Use the prepared customer data to create a CHAID decision-tree model, define the churn target and predictors, and inspect the generated model results.

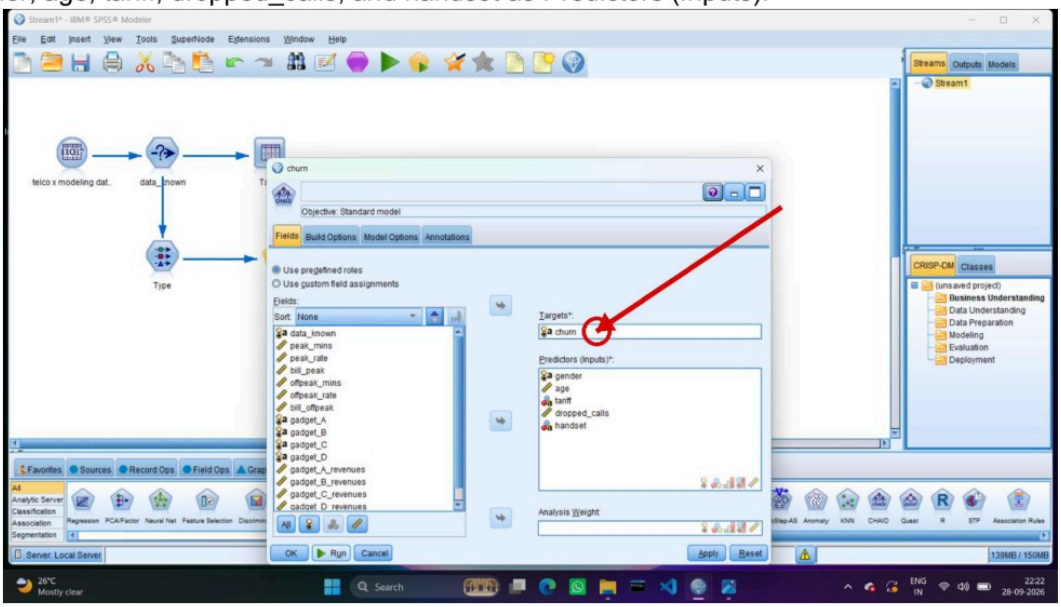
Step 21 – Select the CHAID modelling node

Open the Modeling palette and select the CHAID node. CHAID is used here to construct a decision tree for classifying the customer churn outcome.



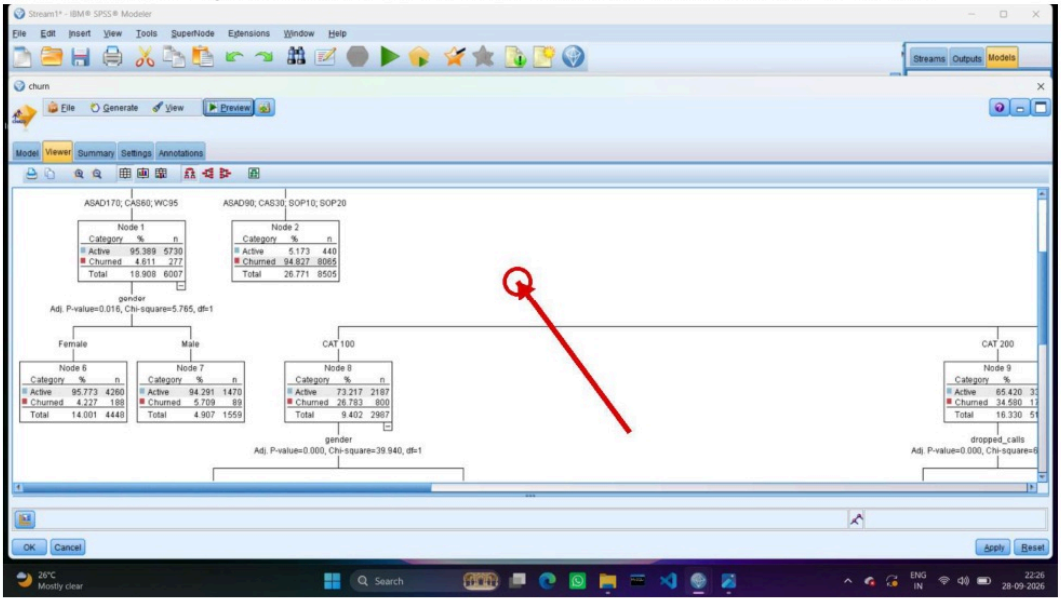
Step 22 – Configure the churn model fields

In the CHAID settings, set churn as the Target and use the displayed customer attributes such as gender, age, tariff, dropped_calls, and handset as Predictors (Inputs).



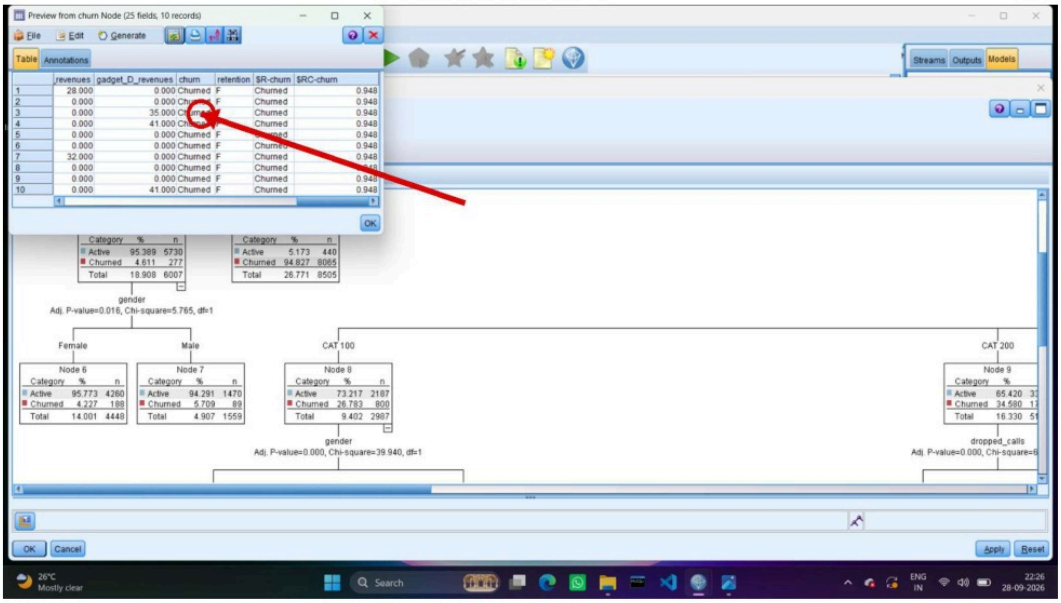
Step 23 – Generate the CHAID model

Run the configured model to generate the decision tree. The resulting tree displays customer groups, predicted categories, record counts, and statistical information for the splits.



Step 24 – Preview the model predictions

Open the churn model preview to inspect the generated prediction fields. The output contains predicted Active/Churned results and associated prediction-confidence information for the displayed records.

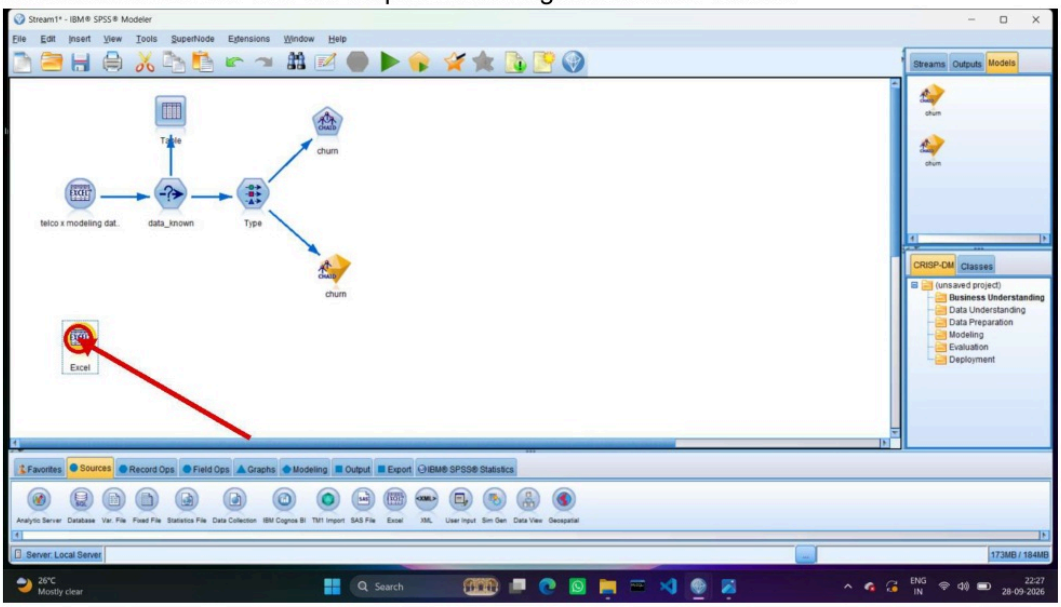


Task 6 – Import and prepare the deployment dataset

What to do: Load the separate deployment workbook and reuse the earlier data-preparation logic so current customer records can be scored using the trained churn model.

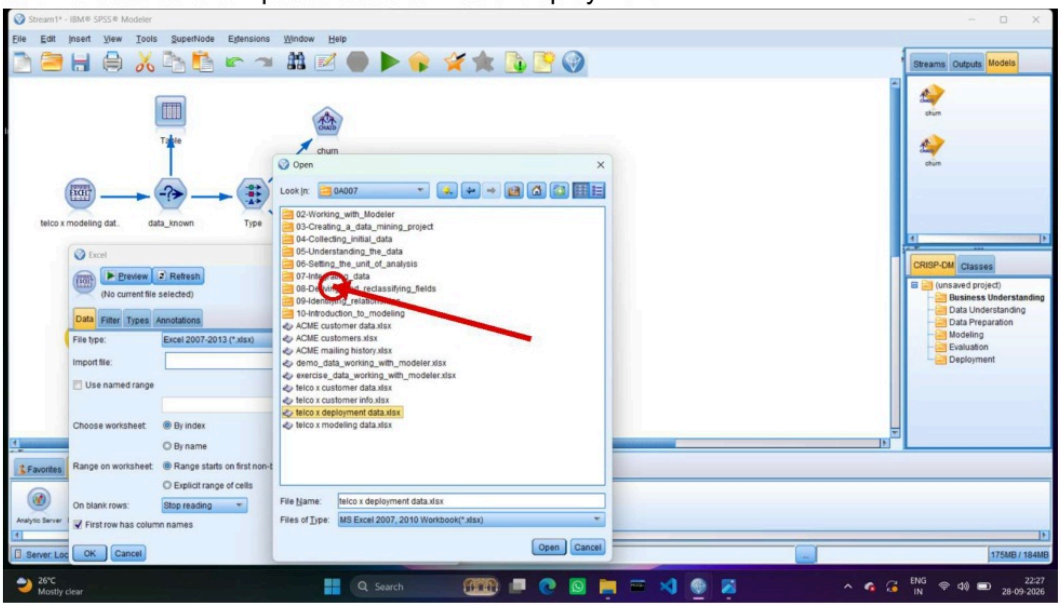
Step 25 – Add the deployment Excel source

Add another Excel source node to the lower part of the stream. This second source represents the current customer records that will be passed through the trained model.



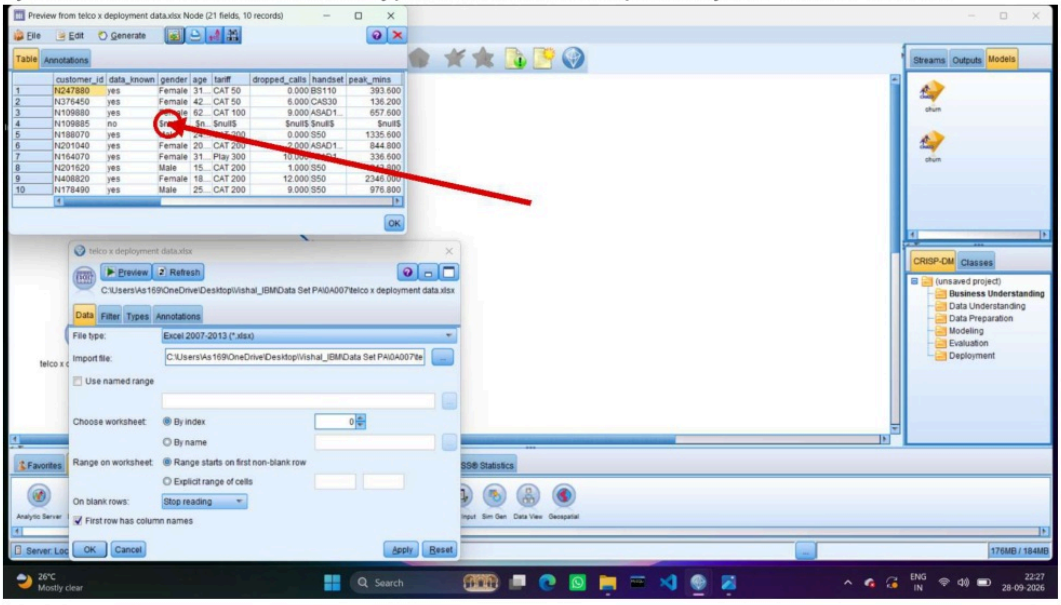
Step 26 – Select the deployment workbook

Open the new Excel node and select telco x deployment data.xlsx from the project files. The workbook is used as the input dataset for model deployment.



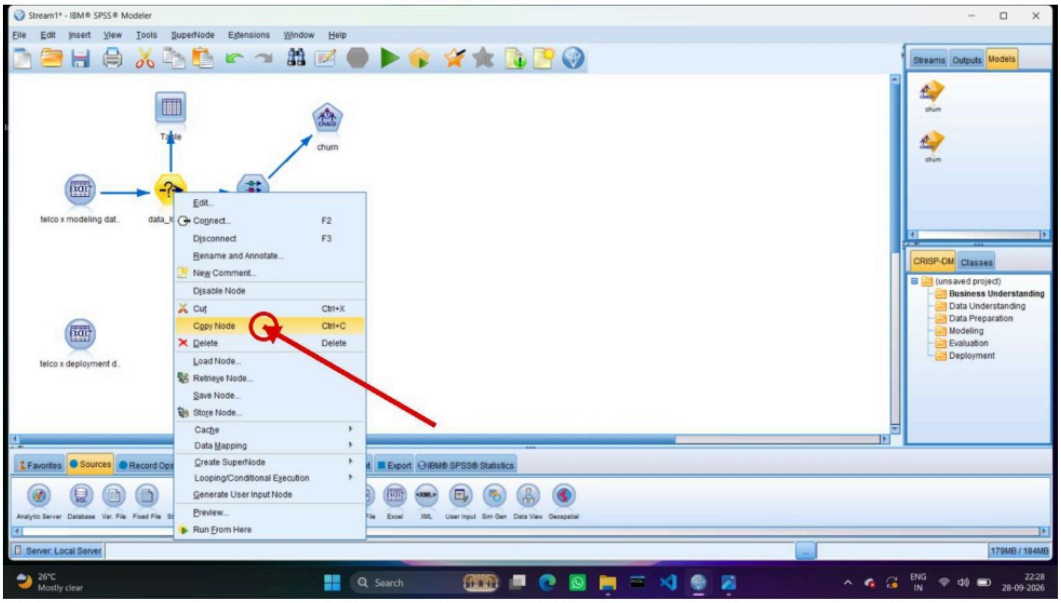
Step 27 – Preview deployment records

Preview the deployment workbook to confirm that the required customer fields are available. The displayed records include the same types of attributes required by the trained model.



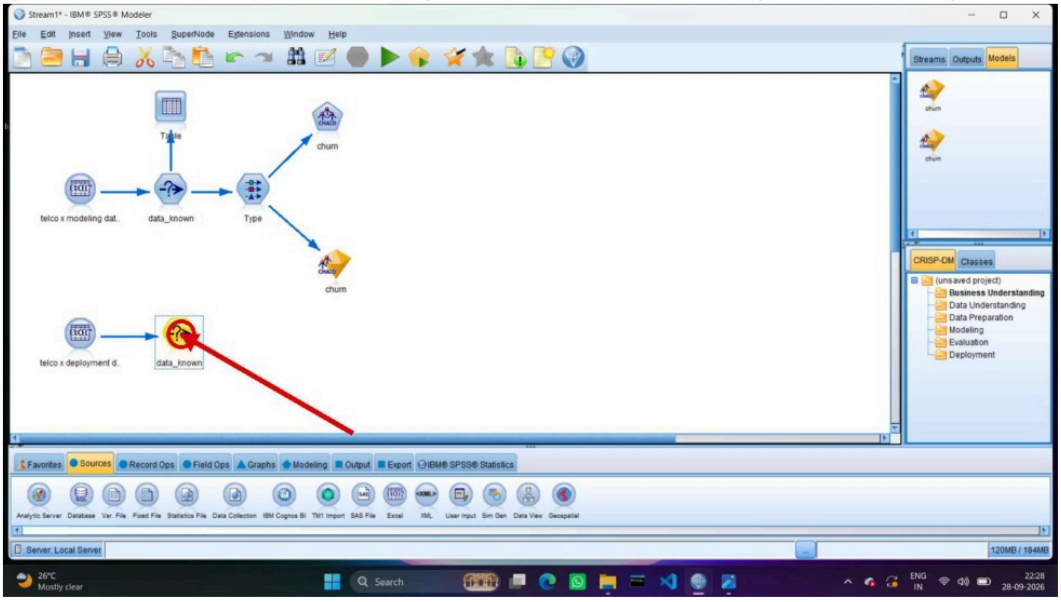
Step 28 – Copy the existing data-preparation node

Right-click the existing data_known node and choose Copy Node. Copying the node allows the same preparation logic to be reused for the deployment data.



Step 29 – Apply the copied preparation node

Place the copied data_known node after the deployment Excel source and connect it. The deployment records can now pass through the same preparation stage before scoring.

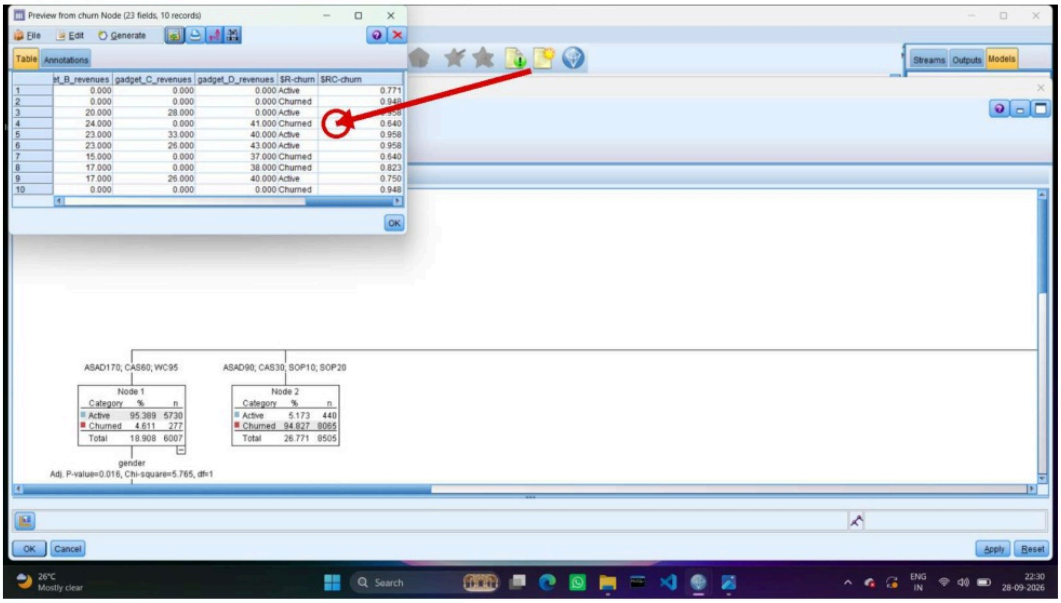


Task 7 – Apply the trained model and create the customer-risk field

What to do: Score the deployment records with the trained churn model, generate prediction output, and give the generated prediction node a meaningful name.

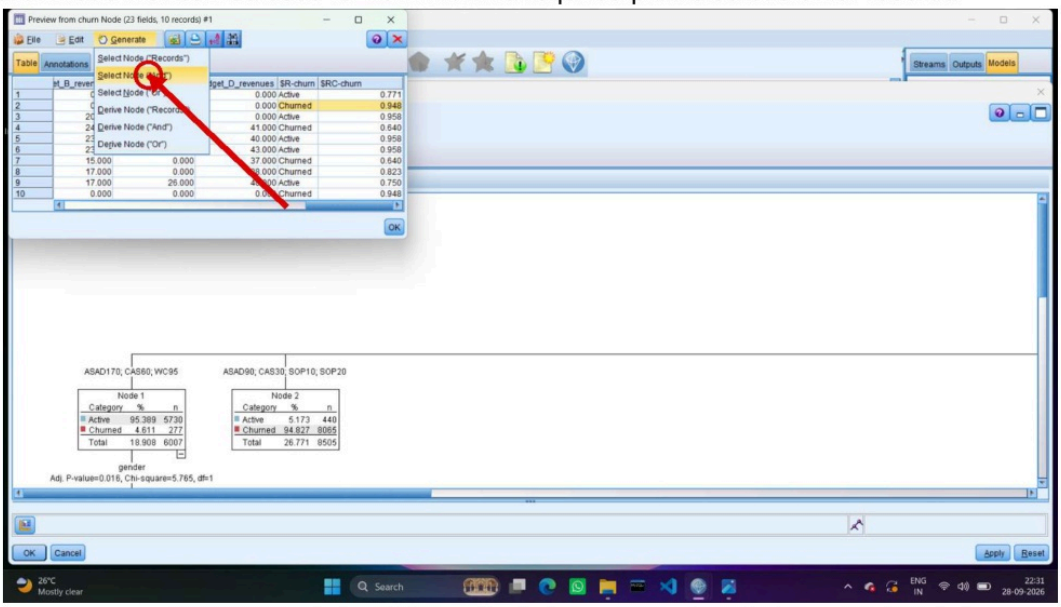
Step 30 – Preview the scored deployment records

Open the churn model output for the deployment branch and review the prediction columns. The output includes fields such as $R-churn and $RC-churn, which represent the predicted category and its confidence.



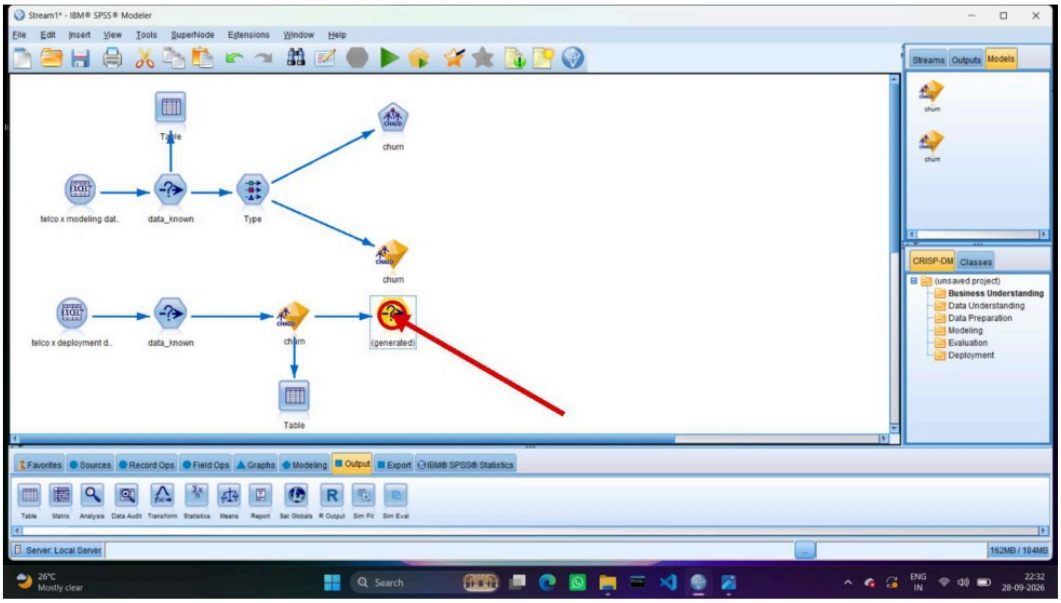
Step 31 – Generate a Select Node from the prediction output

Use the Generate menu in the prediction preview and choose Select Node ('And'). This creates a selection node that can be used to work with the required predicted customer records.



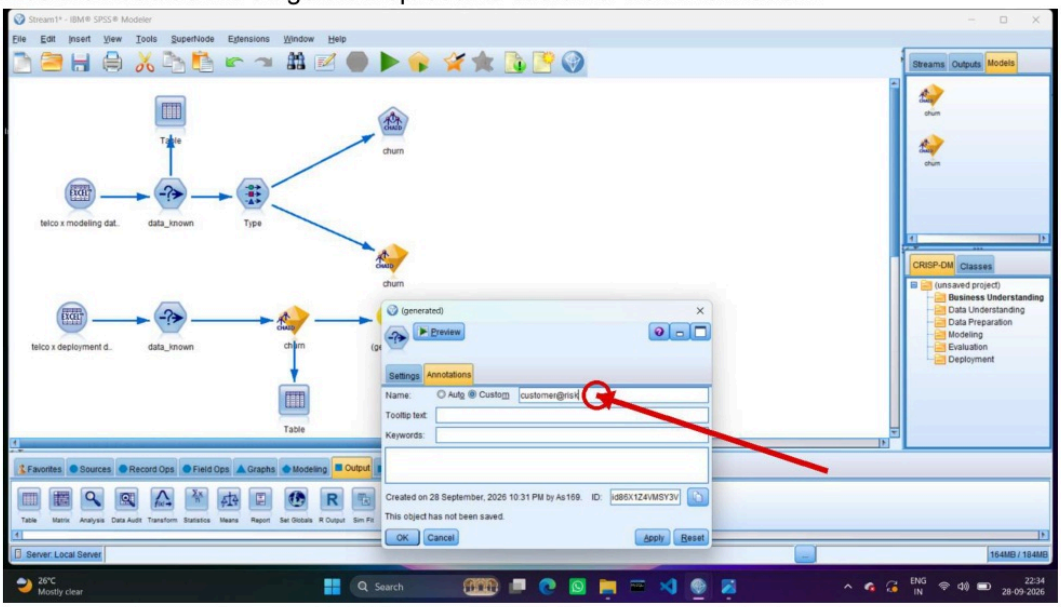
Step 32 – Verify the deployment scoring branch

The deployment branch now connects the deployment source, data preparation, churn model, and generated node. This confirms that the current records are being passed through the predictive workflow.



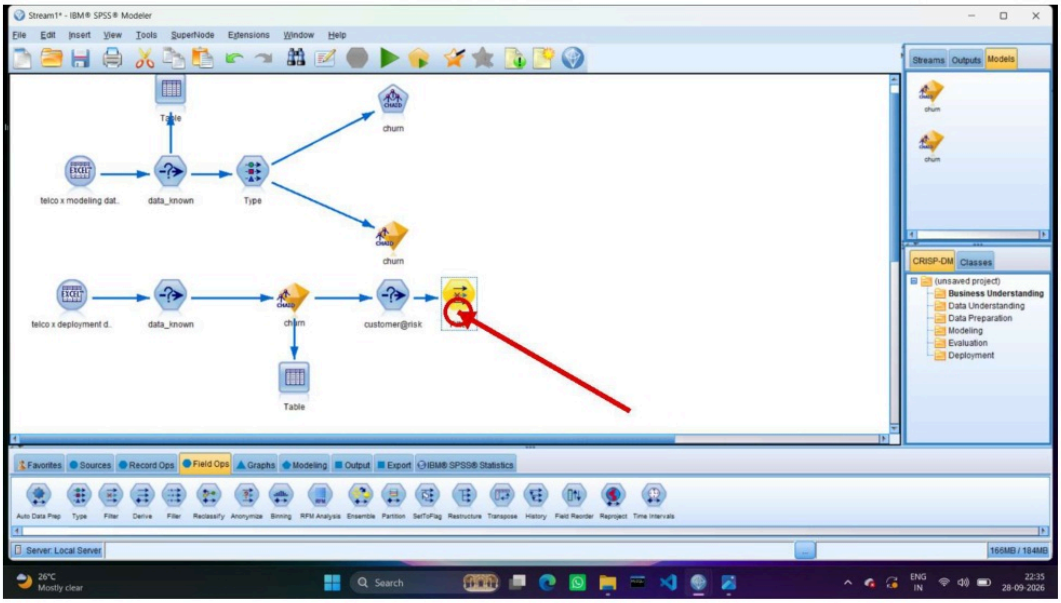
Step 33 – Rename the generated prediction node

Open the generated node settings and assign the custom name customer@risk. This label clearly identifies the node as the stage that represents customer risk information.



Step 34 – Add a Filter node

Place a Filter node after customer@risk and connect it to the deployment branch. The Filter node is used to remove unnecessary fields and retain the information needed for the final output.

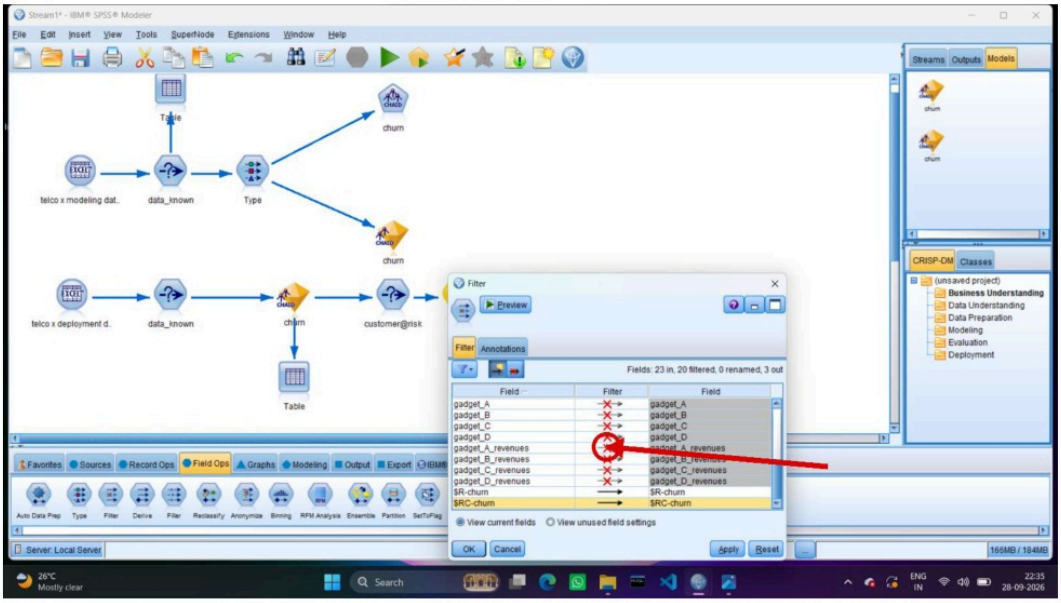


Task 8 – Filter and export the final customer-risk results

What to do: Remove unnecessary fields from the scored data and export the resulting customer-risk information as the final project output.

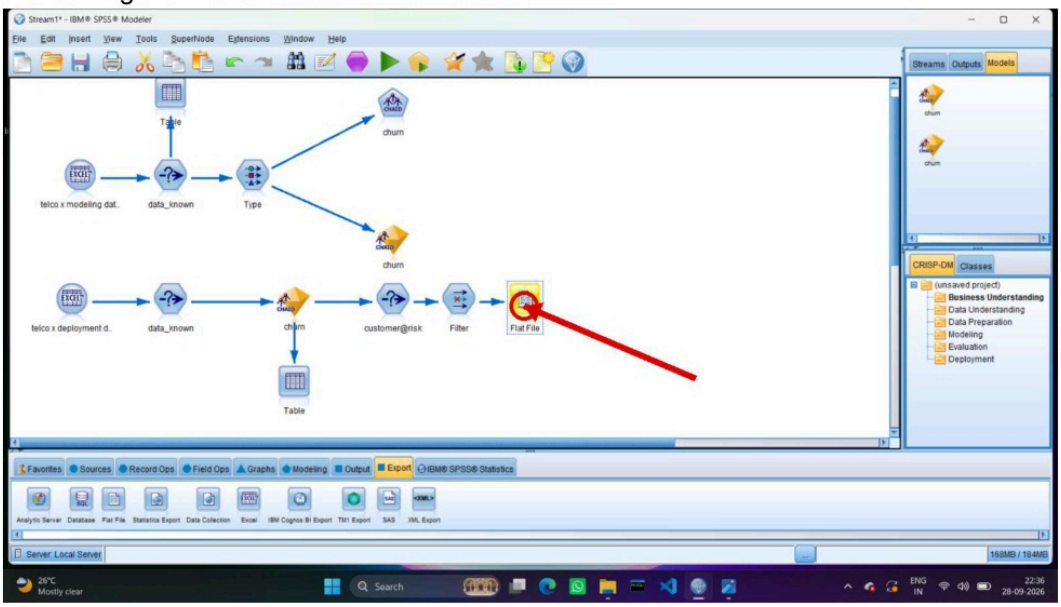
Step 35 – Configure the Filter node

Review the Filter settings and exclude the unnecessary gadget-related fields while retaining the generated churn prediction fields. The screen shows the selected fields being filtered and the retained prediction fields.



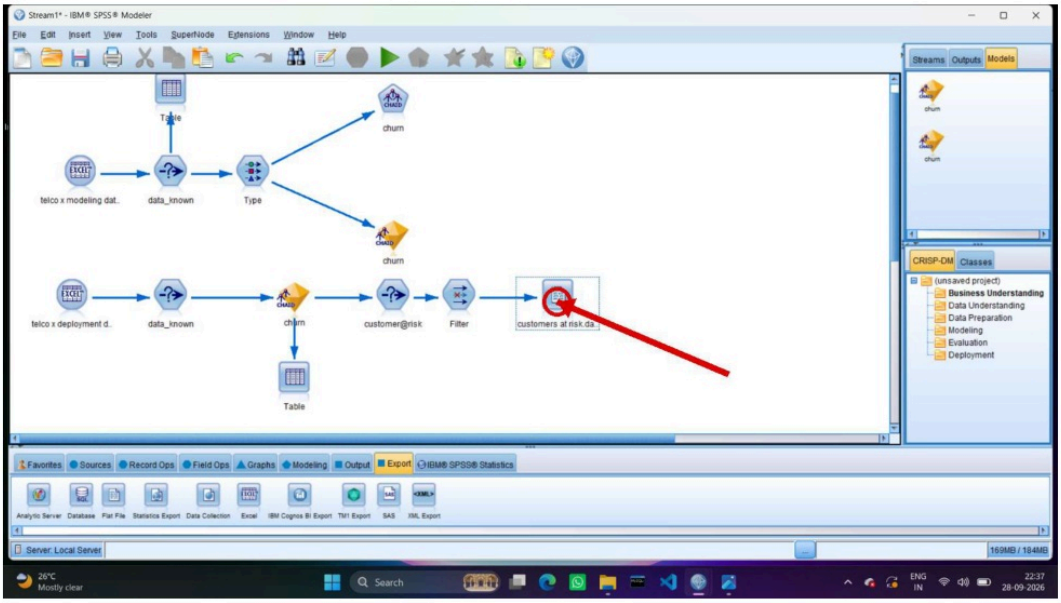
Step 36 – Add a Flat File export node

From the Export palette, add a Flat File node after the Filter node. This node provides the final output stage for saving the filtered customer-risk records.



Step 37 – Verify the completed predictive analytics stream

The final stream connects the deployment data through preparation, CHAID scoring, customer-risk generation, filtering, and file export. This completed flow produces the final customer-risk output for submission.



Project Result

The completed stream imports historical customer data, prepares the fields, trains a CHAID churn model, scores the deployment dataset, generates customer-risk information, filters unnecessary fields, and exports the final results. The workflow therefore covers the complete predictive analytics process from data input to output.